Identifying Structural Typologies of Workplace Aggression in Healthcare: A Text–Validated Analysis of Toscana Nord Ovest Reports

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Abstract. Workplace aggression against healthcare workers poses persistent organizational and safety challenges. We adopt a data-driven approach to identify structural typologies of aggression based on reporting data.

We analyze $n = 2{,}226$ aggression reports from the Local Health Authority (LHA) Toscana Nord Ovest (Italy, 2022–2026). Using binary event descriptors, we apply Partitioning Around Medoids (PAM) under Hamming dissimilarity. Silhouette and bootstrap stability identify a robust two–cluster solution distinguishing patient–driven physically intense from companion–driven predominantly verbal episodes.

Narrative descriptions are analyzed through Term Frequency–Inverse Document Frequency (TF–IDF) representations and sentence embeddings, revealing greater semantic coherence within clusters and distinct lexical profiles aligned with the structural classification. To assess contextual determinants, we train an interpretable CatBoost classifier and examine model interpretability through SHapley Additive exPlanations (SHAP), clarifying the contribution of organizational and temporal variables to typology membership.

These findings demonstrate that structured variables capture meaningful aggression configurations, while narrative data and interpretable machine learning provide complementary validation and insight for monitoring and prevention in healthcare settings.

Keywords: Workplace aggression · clustering · semantic analysis · interpretable machine learning

1 Introduction

Workplace aggression against healthcare workers represents a persistent challenge for healthcare systems, affecting staff safety, organizational functioning,

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and quality of care. In Italy, healthcare and social assistance are among the occupational sectors most exposed to aggressive behaviors, with documented heterogeneity across care settings, professional roles, and temporal contexts [2,4]. These patterns suggest that aggression events may follow recurrent structural configurations rather than occurring as isolated or purely stochastic episodes.

Aggression reporting systems collect structured descriptors (e.g. aggressor characteristics, modality, context) alongside narrative descriptions capturing interactional dynamics. While prior research has primarily focused on prevalence and risk factors [1,3,7], we adopt a typological perspective. Using aggression report data from the LHA Toscana Nord Ovest, we identify structurally coherent configurations of workplace aggression and assess whether narrative descriptions provide independent semantic validation.

We analyze $n = 2{,}226$ anonymized aggression reports (2022–2026) and derive robust structural typologies via distance–based clustering on binary event descriptors. We subsequently assess the semantic coherence of narrative descriptions and examine contextual determinants of typology membership through interpretable machine learning. Together, these steps provide a transparent statistical framework for monitoring workplace aggression and informing prevention strategies.

2 Data Source

Each workplace aggression event is recorded as a structured report including contextual variables (ward, location, time slot), victim characteristics (profession, gender, age), and event phenomenology (e.g. aggression modality, aggressor type, use of objects). Additionally, all reports contain a free–text narrative description (mean length = 430 characters).

The dataset comprises $n = 2{,}226$ anonymized aggression reports from April 2022 to February 2026, completed by the workers of the LHA Toscana Nord Ovest. The majority of events involve female workers (75%) and nurses (48%). Aggression modalities are not mutually exclusive, with verbal and physical aggression frequently co–occurring.

3 Methods

The primary objective was to identify intrinsic structural configurations of aggression events based exclusively on event–level characteristics. Variables describing organizational setting (e.g. ward, location), temporal seasonality, and individual attributes (e.g. age) were excluded from the clustering procedure to ensure that resulting typologies reflect the interactional structure of aggression rather than contextual prevalence.

The structured feature set included binary indicators of aggressor type, aggression modality, interactional dynamics, and victim gender. Each event was encoded as a binary vector in $\{0,1\}^p$, where $p = 12$ denotes the dimensionality of the structured feature space.

Pairwise dissimilarities were computed using the normalized Hamming distance, defined as the proportion of mismatched components between two binary vectors. This distance defines a proper metric on $\{0,1\}^p$, satisfying non-negativity, symmetry, and the triangle inequality, and is therefore well suited for symmetric binary attributes.

Clustering was performed directly on the dissimilarity matrix using PAM, a robust method that operates in arbitrary metric spaces and identifies representative observations (medoids) as cluster centers [5].

3.1 Number of Clusters and Robustness Assessment

The number of clusters K was selected using two complementary criteria: average silhouette width and stability under bootstrap resampling measured by the Adjusted Rand Index (ARI). Silhouette evaluates separation and cohesion directly on the dissimilarity matrix, while ARI quantifies agreement between clustering across resampled datasets.

We adopted a plateau rule, selecting the most stable solution within 95% of the maximum silhouette value. Across tested configurations (from $K = 2$ to $K = 9$), a two–cluster solution achieved the highest separation (silhouette ≈ 0.41) and high stability (ARI ≈ 0.75), indicating a dominant bipartite structure.

To assess sensitivity to clustering specification, we replicated the analysis using alternative configurations, including hierarchical clustering, PCA followed by k-means, and multidimensional scaling (MDS) combined with k-medoids. The resulting partitions showed substantial agreement (ARI ≈ 0.77), confirming the robustness of the identified macro-structure.

3.2 Modeling Typology Determinants

To examine the extent to which organizational and contextual factors predict typology, we frame cluster membership labels as an outcome variable and consider supervised classification models. In particular, we employ interpretable machine learning techniques to estimate the probability of belonging to each typology as a function of contextual variables (e.g. ward, time slot, professional role).

Model interpretability is assessed using SHAP, which quantifies the marginal contribution of each predictor to individual and global predictions. This step allows us to distinguish between structural event configurations and their organizational determinants, thereby linking typology identification with actionable risk factors.

3.3 Post–hoc Semantic Validation

Narrative descriptions were not used to derive the structural typologies but were analyzed post–hoc to evaluate semantic coherence and lexical differentiation across clusters. Texts were minimally normalized and represented using TF–IDF

weighting on word–level unigrams and bigrams with Italian stopword removal [8]. The vocabulary was capped at 5,000 features.

Cosine similarity was used to quantify semantic proximity between documents in the vector space model. For each cluster, we computed (i) the mean within–cluster cosine similarity and (ii) the mean cosine similarity to documents in other clusters, and examined their difference as a measure of semantic concentration [9]. Since cosine similarity is bounded in $[-1, 1]$, no additional scaling was applied.

To complement the sparse TF–IDF representation, we computed 768–dimensional sentence–level embeddings for each description using the transformer–based model `nomic--embed--text--v2--moe` [6], deployed locally via Ollama to ensure compliance with data protection constraints. The model supports a context window of up to 512 tokens and produces fixed–length dense vector representations. These embeddings were used exclusively for post–hoc validation and not for cluster construction. Cosine similarity in the embedding space was employed to assess intra– and inter–cluster semantic coherence.

Cluster–specific lexical markers were identified through a contrastive TF–IDF approach, defined as the difference between the mean TF–IDF weight of each term within a cluster and its mean weight in the remaining clusters. In addition, central descriptions were identified as those minimizing cosine distance from the cluster centroid, supporting qualitative interpretation of the typologies.

As a robustness check, clustering was repeated using semantic embeddings alone, yielding negligible agreement with the structural solution (ARI ≈ 0.01), confirming that textual representations did not drive the primary partition and thus supporting the post–hoc validation framework.

4 Results

Silhouettes and bootstrap stability identified a two–cluster solution as the most robust configuration, see Fig. 1 Panel A. The optimal partition achieved a silhouette value ≈ 0.41 and ARI ≈ 0.75, indicating strong separation and structural reproducibility. Alternative dissimilarity specifications yielded substantially similar partitions (ARI ≈ 0.77), supporting the stability of the bipartite structure.

Chi–square tests confirmed marked structural differences between clusters. All reported associations were statistically significant at $p < 0.001$. Aggressor type showed strong association with cluster membership (Cramér's $V = 0.95$ for patient aggressors [*agg_patient*]; $V = 0.67$ for companions [*agg_comp*]). Physical aggression [*physical*] ($V = 0.32$), and use of objects [*use_of_obj*] ($V = 0.20$) were also significantly associated, confirming the interpretation of a physically intense patient–driven cluster and a predominantly verbal companion–driven cluster.

Variables excluded from the clustering procedure were examined post–hoc. Temporal variables showed small but significant associations (time slot $V = 0.09$). Age did not differ significantly between clusters (Mann–Whitney $p = 0.097$). Among short-term consequences descriptors, only interruption of work activities differed ($V = 0.10$), whereas emergency access and prognosis did not.

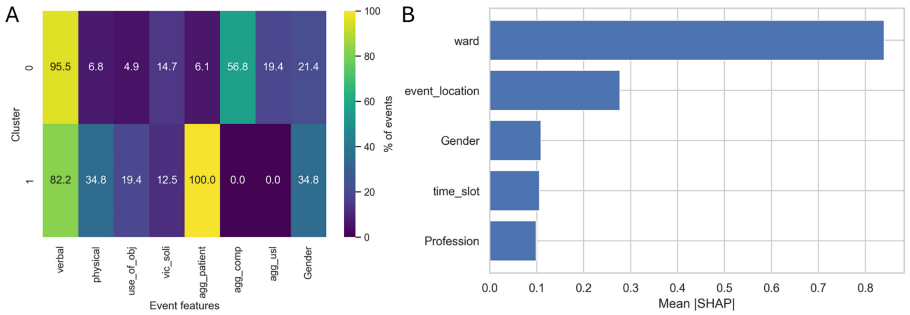


Fig. 1. Panel A: Prevalence of key structural indicators across the two identified typologies. Panel B: Global SHAP importance for the top predictors of typologies.

Semantic coherence metrics are reported in Table 1. In the TF–IDF space, both clusters showed higher within–cluster similarity than between–cluster similarity. More pronounced differentiation emerged in the embedding space, where cluster 1 exhibited a larger intra–inter similarity gap, indicating stronger semantic compactness. Contrastive TF–IDF analysis revealed distinct lexical profiles across the two clusters. Cluster 0 exhibited over–representation of relational and family–related terms, while Cluster 1 was characterized by care–context and violence–related expressions. These patterns closely mirror the structural typology identified through binary event descriptors.

Table 1. Semantic coherence of structural typologies: mean cosine similarity within clusters and between clusters in TF–IDF and embedding spaces.

Cluster	Intra – Inter TF–IDF	Intra – Inter Emb.	Top contrastive terms
0	0.025–0.019	0.419–0.409	*parente, madre, figlia, collega*
1	0.022–0.019	0.435–0.409	*paziente, terapia, calci, pugni*

To further examine contextual determinants of typologies, we trained an interpretable CatBoost classifier using organizational and temporal variables (ward, location, profession, time slot, age). The model achieved ROC–AUC = 0.816 and balanced accuracy = 0.740, indicating solid predictability of cluster labels from contextual factors alone.

SHAP analysis identified ward as the most influential predictor of cluster membership (see Fig. 1, Panel B), which was also strongly associated with typology (Cramér's $V = 0.49$). Physically intense, patient–driven episodes were highly prevalent in psychiatric and addiction services (often exceeding 85% of events), whereas companion–driven verbal episodes predominated in residential and community settings (around 70–95%). Professional roles displayed differentiated exposure patterns, with psychologists and security staff relatively more involved in physically intense configurations.

5 Conclusions

We identify a robust bipartite structural typology of workplace aggression, patient–driven, physically intense episodes and companion–driven, predominantly verbal events, using distance–based clustering on binary event descriptors, supported by semantic validation and interpretable modeling. The consistency between structural patterns and narrative content strengthens confidence in the stability and interpretability of the proposed framework.

Future research should incorporate measures of aggression severity and further examine the interaction between structural typologies, contextual determinants, and textual features, which provide an analytical perspective orthogonal to the structural dimension. Such developments would enhance both methodological depth and practical relevance for monitoring workplace aggression and informing targeted prevention strategies.

Data Availability Statement. The authors had access to anonymized reports of aggression events recorded between 2022 and 2026 by the LHA Toscana Nord Ovest. Access to these data was granted under a formal agreement between the Management and Health Laboratory of the Sant'Anna School of Advanced Studies and the LHA Toscana Nord Ovest, aimed at supporting the monitoring and prevention of violence against healthcare workers.

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